EMR-Diff: Edge-aware Multimodal Residual Diffusion Model for Hyperspectral Image Super-resolution

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Abstract

Hardware constraints make it challenging to simultaneously acquire hyperspectral images (HSIs) with both high spatial and high spectral resolutions. A promising solution is to fuse low-resolution HSI (LR-HSI) with high-resolution multispectral images (HR-MSI) to generate high-resolution HSI (HR-HSI). Recently, diffusion models have introduced possibilities for HSI super-resolution, but suffer from low-efficiency sampling, detail-limited generation, and insufficient denoising. To address these issues, we propose an Edge-aware Multimodal Residual Diffusion Model (EMR-Diff). Specifically, multimodal residual mechanism is introduced to facilitate efficient information transfer among HR-MSI, LR-HSI, and HR-HSI, significantly improving the fusion efficiency. Edge-aware noise strategy is designed by exploiting the edge information of HR-MSI, which guides the model to prioritize high-frequency detail reconstruction by applying stronger noise perturbations to edge regions. In addition, we propose a Bilateral Attention Fusion UNet and design a multi-scale supervision mechanism to enable progressive reconstruction and collaborative optimization of spectral and spatial features. Extensive experiments demonstrate that our method achieves superior performance over existing approaches in both quantitative metrics and visual quality. Our code is available at: https://github.com/luocz55/EMR-Diff.

1. Introduction

Hyperspectral image (HSI) continuously samples across the visible to short-wave infrared range, revealing fine spectral features of objects [2]. Consequently, it is widely applied in fields such as environmental monitoring [33], geological exploration [36], military [35], and food safety [10]. However, achieving both high spectral and spatial resolution remains challenging due to physical sensor limitations. To overcome this, researchers are developing computational imaging and software algorithms to enhance spatial resolution.

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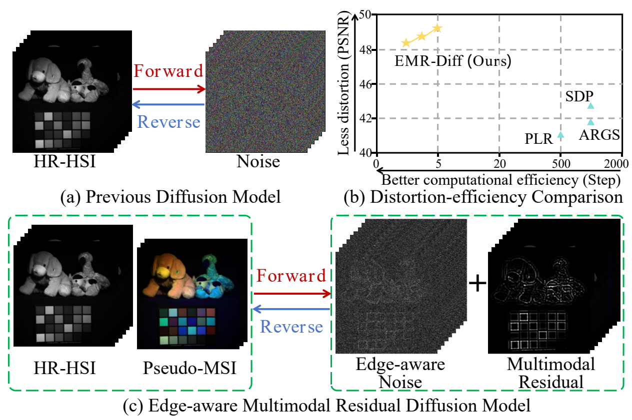


Figure 1. Comparison of different diffusion models. With edge-aware noise and multimodal residual, EMR-Diff significantly improves the performance and efficiency.

The current HSI super-resolution technology is divided mainly into two categories, *i.e.*, single-image super-resolution [23, 24, 50, 59] and multi-image fusion-based super-resolution [7, 41, 52, 57]. The reconstruction of a single low-resolution HSI (LR-HSI) faces fundamental challenges due to severe information deficiencies, often resulting in distortion issues such as ghosting. In contrast, fusion-based methods can achieve significantly better imaging quality by introducing high-resolution multispectral images (HR-MSI) as a supplement to spatial details.

In the development of multi-image fusion-based HSI super-resolution, early traditional methods [6, 8, 58] are mainly based on mathematical models and manually designed prior knowledge for fusion. However, these methods have weak adaptability to complex real-world scenes, resulting in limited imaging effects. With the rise of deep learning technology, research has gradually turned to end-to-end learning through neural networks. At present, deep learning-based multi-image fusion for HSI super-resolution is primarily categorized into unsupervised and supervised methods. Unsupervised methods [11, 38, 47, 60] do not require supervised signals and have strong generalization ability, but their imaging quality still faces significant constraints in some complex scenarios. For supervised methods [5, 16, 17, 26, 28, 34, 43], with the release of more hyperspectral datasets, the problem of data scarcity has been alleviated. However, existing methods still exhibit insufficient generalization ability in cross-scenario applications.

Recently, diffusion model [15] has gradually become a new direction for HSI super-resolution due to its powerful image generation capabilities and good interpretability. However, diffusion models face some challenges, as shown in Fig. 1. **1) Low-efficiency sampling.** It typically requires a large number of iterative steps during sampling, resulting in low inference efficiency. **2) Detail-limited generation.** The pure Gaussian noise employed is difficult to effectively focus on the restoration of high-frequency details in HSI super-resolution tasks, thereby limiting the quality of the generated images. **3) Insufficient denoising.** Conventional UNet architectures suffer from issues such as information fusion interference, blind upsampling processes, and insufficient utilization of observed information when reconstructing HSI, which also constrain the further improvement of the model's overall performance.

In this work, we propose an Edge-aware Multimodal Residual Diffusion model (EMR-Diff) for fusion-based HSI super-resolution. Specifically, the concatenation of a high-resolution HSI (HR-HSI) and its pseudo MSI serves as the starting point of the Markov chain, while the concatenation of LR-HSI and HR-MSI serves as the end point. The number of diffusion steps is shortened by tens of times, via gradually transferring multimodal residual between the starting point and the end point. Then, to reconstruct perceptually critical high-frequency structures, an edge-aware noise control strategy is proposed. It converts the edge information of HR-MSI into noise weights, applying less noise to flat areas and more noise to edge regions, thereby guiding the model to focus on the generation of high-frequency structures. Finally, we propose the Bilateral Attention Fusion UNet (BAF-UNet), which is divided into a denoising path and a guiding path to avoid feature confusion caused by early fusion. Each path uses a Multi-scale Group Attention Block (MSGAB), which integrates grouped convolution, spatial channel attention, and adaptive feature fusion to comprehensively capture different types of feature. During the upsampling process, we innovatively use the downsampled HR-HSI as the supervision target for each upsampling stage, thereby simulating a progressive reconstruction from low resolution to high resolution, and solving the problem of upsampling blindness. The experimental results show that EMR-Diff outperforms state-of-the-art methods in both quantitative metrics and qualitative visualization.

The main contributions are summarized as follows:

- We propose an efficient Edge-aware Multimodal Residual Diffusion Model, which reduces the sampling cost by migrating multimodal residual between HR-MSI, LR-HSI, and HR-HSI. During inference, the required diffusion steps are reduced to dozens by adding the multimodal residual between the end and start points.
- We utilize the edge information of HR-MSI to modulate noise in the diffusion process. By applying stronger noise in the high-frequency edge regions, the denoising network can focus on the recovery of image details, significantly enhancing spatial fidelity of reconstructed images.
- We present a denoising architecture BAF-UNet with three complementary components, *i.e.*, modality-specific dual-path design, multi-scale supervision, and MSGAB with adaptive fusion. It significantly enhances the model's capacity to produce high-quality reconstructions.

2. Related Work

2.1. Traditional HSI Super-resolution

Traditional HSI super-resolution can be roughly divided into four categories, including component replacement method [14], multi-resolution analysis method [18], subspace based method [45], and tensor based method [32]. For example, Wei *et al.* [39] learn a dictionary and sparse support from observed images, and use an alternating optimization strategy for image fusion. Zhang *et al.* [49] propose a coupled tensor decomposition method for image fusion by utilizing the characteristics of canonical multivariate decomposition. Dong *et al.* [9] propose a non-negative dictionary learning algorithm based on block coordinate descent to estimate sparse encoding of HSIs and obtain high-resolution results. Zhang *et al.* [56] propose a fusion method based on three-dimensional wavelets for the characteristics of MSI and HSI data volume and three-dimensional feature analysis. Moeller *et al.* [29] propose a wavelet variational method for fusing high-resolution images and HSIs with any number of bands. However, traditional methods are limited by linear mathematical models and manually designed priors, resulting in the issues of spectral distortion and insufficient spatial details.

2.2. Deep Learning HSI Super-resolution

Deep learning methods [12, 13, 19, 21, 22, 51, 53, 54, 61] implicitly model complex nonlinear mappings from degraded to target space through their deep structures. Unlike traditional approaches, Deep HSI super-resolution methods do not rely on any preset mixing model, but directly learn the underlying physical mechanisms of spectral mixing and unmixing through data-driven approaches. These methods

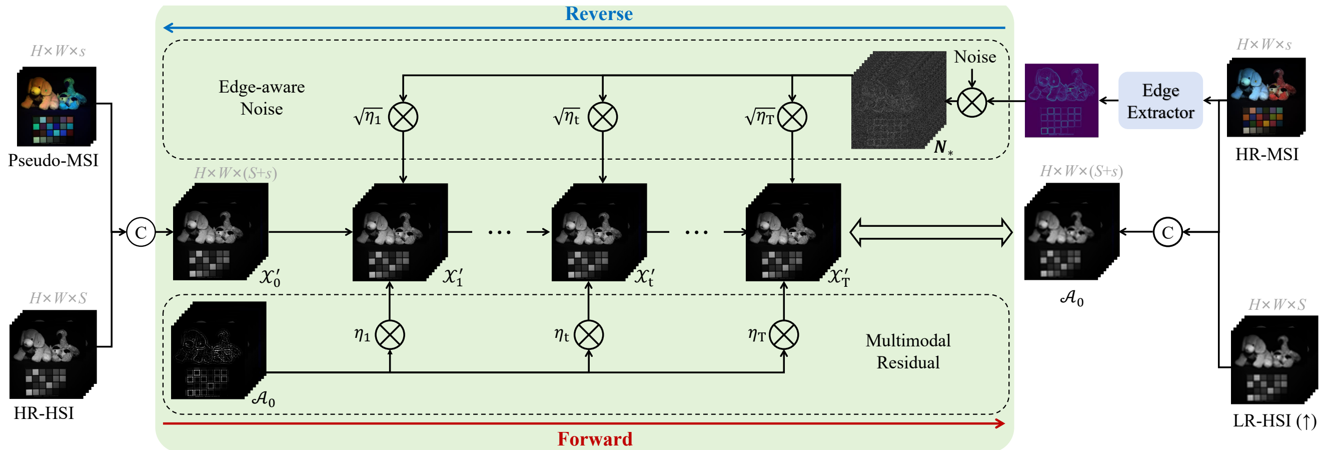


Figure 2. Architecture of the proposed EMR-Diff for HSI super-resolution. The multimodal residual $\mathcal{E}_0$ is obtained by subtracting $\mathcal{A}_0$ from $\mathcal{X}_0'$, and edge-aware noise $\boldsymbol{N}_*$ is generated by modulating Gaussian noise with the edge information of HR-MSI. During the forward diffusion process, the starting point $\mathcal{X}_0'$ is gradually injected with $\mathcal{E}_0$ and $\boldsymbol{N}_*$, both controlled by a monotonically increasing sequence η_t.

can be broadly categorized into supervised and unsupervised methods. Supervised methods rely on paired training data to learn the mapping function. Xie *et al.* [42] construct a HSI fusion model by expanding the proximal gradient algorithm. Huang *et al.* [34] solve the problem of deep HSI fusion by jointly utilizing spatial spectral regularization and physical imaging models. Zhang *et al.* [55] propose an interpretable spatial spectral reconstruction network based on cross-modal information fusion. Besides, unsupervised methods only use the observation image to constrain the model output image. Li *et al.* [20] propose an enhanced deep image prior method that accurately simulates complex HSI priors. Cao *et al.* [3] introduce an unsupervised hybrid network for blind HSI fusion. Despite the encouraging results achieved by the aforementioned supervised and unsupervised deep learning methods, they often fall short in generating realistic high-frequency details. It has motivated the pursuit of more powerful image generation models.

2.3. HSI Super-resolution Based on Diffusion Model

In recent years, image generation models such as diffusion models, autoregressive models, and generative adversarial networks (GANs) have been widely used in the field of HSI fusion due to their powerful generation capabilities. For example, Liu *et al.* [25] propose a HSI fusion method that utilizes a spatial autoregressive model for internal structure guidance. Wang *et al.* [37] present an HSI fusion method called HyperGAN based on GANs. Rui *et al.* [31] employ a low rank diffusion model for HSI pan-sharpening. Wu *et al.* [40] propose an HSI fusion approach with a conditional diffusion model. Qu *et al.* [30] propose a spatial-spectral-bilateral cyclic diffusion framework for HSI fusion.

The diffusion models have become the current mainstream architecture. However, when applied to HSI super-resolution, ordinary diffusion models face several key challenges, including low-efficiency sampling, detail-limited generation, and insufficient denoising. In this work, we propose an EMR-Diff method, which introduces a multimodal residual transfer mechanism, edge-aware noise strategy, and a specialized BAF-UNet architecture to enable efficient and high-quality HSI super-resolution.

3. EMR-Diff

Although some previous diffusion-based HSI super-resolution works [30, 40] have achieved good results, these methods directly apply the standard framework (e.g., DDPM [15]), which has the shortcomings of low-efficiency sampling, detail-limited generation, and insufficient denoising. To solve these problem, we propose an EMR-Diff method, whose main components are multimodal residual mechanism, edge-aware noise, and BAF-UNet. The structure diagram of EMR-Diff is shown in Fig. 2.

Notation. In general, the relationship between LR-HSI, HR-MSI, and HR-HSI can be expressed as

$$\mathcal{Y} = \mathcal{D}(\mathcal{B}(\mathcal{X})), \quad \mathcal{Z} = \mathbf{R}\mathcal{X}, \tag{1}$$

where $\mathcal{X} \in \mathbb{R}^{H\times W\times S}$, $\mathcal{Y} \in \mathbb{R}^{h\times w\times S}$, and $\mathcal{Z}$ represents the tensor form of HR-HSI, LR-HSI, and HR-MSI, respectively. $\mathcal{B}$ and $\mathcal{D}$ are blurring and downsampling operations, $\mathbf{R}$ represents the spectral response function of HR-MSI.

3.1. Multimodal Residual

Inspired by Resshift [48], focusing on natural image super-resolution with unimodal residual, EMR-Diff extends the concept to multimodal image fusion by explicitly modeling the residual between HR-HSI, LR-HSI, and HR-MSI. It can capture not only spatial discrepancies but also spectral inconsistencies, which are critical in HSI reconstruction.

Forward Process. The forward process of the basic diffusion model is to gradually add Gaussian noise to an original image, eventually completely destroying it into a pure noise image. EMR-Diff improves to fully utilize the information of LR-HSI and HR-MSI and shorten the time step. First, to ensure the smoothness of the spectrum, we intercept the $(1, ..., s)$ band of HR-HSI as Pseudo-MSI, and concatenate it with HR-HSI to get the starting point $\mathcal{X}_0'$ of the forward process. Secondly, we use bicubic interpolation to upsample LR-HSI by a factor to obtain $\mathcal{Y}_\uparrow$, and then concatenate $\mathcal{Y}_\uparrow$ with HR-MSI $\mathcal{Z}$ to obtain $\mathcal{A}_0$, this process follows

$$\mathcal{A}_0 = \mathcal{Y}_\uparrow \copyright \mathcal{Z}, \tag{2}$$

where © is concatenation in the channel dimension. Next, we represent multimodal residual between $\mathcal{X}_0'$ and $\mathcal{A}_0$ as

$$\mathcal{E}_0 = \mathcal{A}_0 - \mathcal{X}_0', \tag{3}$$

where $\mathcal{E}_0$ represents the joint discrepancy between the HR-HSI and the combination of the LR-HSI and HR-MSI. In essence, it summarizes the spatial detail loss caused by downsampling between LR-HSI and HR-HSI, as well as the spectral information gap between HR-MSI and Pseudo-MSI. Then, by propagating the multimodal residual through the Markov chain, we effectively guide the diffusion model to reconstruct the missing spatial structures and spectral properties, rather than relying solely on a generic denoising process. We gradually transfer $\mathcal{E}_0$ from $\mathcal{X}_0'$ to $\mathcal{X}_t'$ through the Markov chain, which can be expressed as

$$\mathcal{X}_t' = \mathcal{X}_{t-1}' + \alpha_t \mathcal{E}_0 + \kappa\sqrt{\alpha_t}\boldsymbol{N}_*, \tag{4}$$
$$\mathcal{X}_t' = \mathcal{X}_0' + \eta_t \mathcal{E}_0 + \kappa\sqrt{\eta_t}\boldsymbol{N}_*, \tag{5}$$

where κ is the parameter that regulates the intensity of noise, we set it to 1. $\alpha_t = \eta_t - \eta_{t-1}$, η_{t-1} is a monotonically increasing sequence. $t = 1, 2, \ldots, T$. When $t = 1$, η_t approaches 0, $\mathcal{X}_1'$ is similar to $\mathcal{X}_0'$. When $t = T$, η_t approaches 1, $\mathcal{X}_T'$ is similar to $\mathcal{A}_0$ plus edge-aware noise $\boldsymbol{N}_*$, which will be detailed in the next section. In this way, the multimodal residual is added to the Markov chain, which can reduce the number of sampling steps.

Reverse Process. The reverse process of EMR-Diff starts from $\mathcal{A}_0$ and edge-aware noise and recovers $\mathcal{X}_0'$ by gradually denoising and back-transferring the multimodal residual. The posterior distribution can be expressed as

$$\mathcal{X}_{t-1}' = \frac{\eta_{t-1}}{\eta_t}\mathcal{X}_t' + \frac{\alpha_t}{\eta_t} f_{\boldsymbol{\theta}}(\mathcal{X}_t', \mathcal{A}_0, t) + \kappa\sqrt{\frac{\eta_{t-1}}{\eta_t}\alpha_t}\boldsymbol{N}_*, \tag{6}$$

where the denoising network $f_{\boldsymbol{\theta}}$ is used to predict $\mathcal{X}_0'$.

3.2. Edge-aware Noise

In EMR-Diff, the denoising network $f_{\boldsymbol{\theta}}$ not only needs to learn the distribution characteristics of noise, but also needs

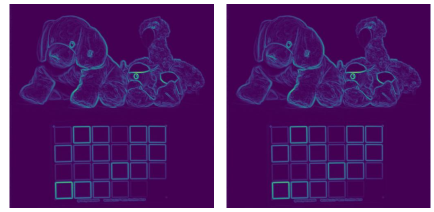


Figure 3. Edge similarity of HR-HSI and MR-HSI.

to model the migration law of multimodal residual. Since multimodal residual is mainly concentrated in the high-frequency areas of the image, especially the edge structure part, its amplitude is usually the most significant. Therefore, we constructed an edge-aware noise to replace the traditional Gaussian noise, and the model can focus more on the high-frequency edge structure information of the image during the denoising process. This method is inspired by the high similarity of the high-frequency edge information in HR-HSI and HR-MSI, as shown in Fig. 3. Sobel operator is used to extract edge information of HR-MSI. First, we convert the HR-MSI into a grayscale image $\boldsymbol{P}$ by band averaging. Then, predefined Sobel operators $\boldsymbol{C}_x$ and $\boldsymbol{C}_y$ are used to compute gradient components in the horizontal and vertical directions, respectively. It can be expressed as

$$\boldsymbol{G}_x = \boldsymbol{C}_x * \boldsymbol{P}, \boldsymbol{G}_y = \boldsymbol{C}_y * \boldsymbol{P}, \tag{7}$$

where $\boldsymbol{C}_x$ and $\boldsymbol{C}_y$ are as follows

$$\boldsymbol{C}_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \boldsymbol{C}_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{8}$$

Next, we use $\boldsymbol{G}_x$ and $\boldsymbol{G}_y$ to calculate the gradient magnitude as the edge strength, which can be represented as

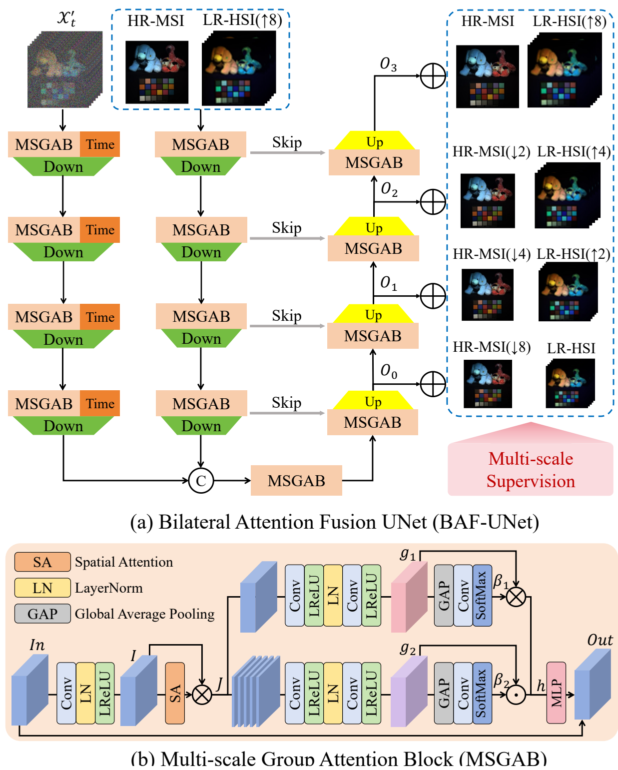
$$\boldsymbol{M} = \sqrt{\boldsymbol{G}_x^2 + \boldsymbol{G}_y^2 + \epsilon}, \epsilon = 10^{-8}. \tag{9}$$

Finally, we normalize $\boldsymbol{M}$ to obtain the weight $\boldsymbol{W}$, and multiply it by pure Gaussian noise $\boldsymbol{N}$ to obtain edge-aware noise $\boldsymbol{N}_*$. The process can be expressed as

$$\boldsymbol{N}_* = \boldsymbol{N} \cdot \boldsymbol{W} = \boldsymbol{N} \cdot norm(\boldsymbol{M}). \tag{10}$$

3.3. Bilateral Attention Fusion UNet

For the reverse process of the diffusion model, the denoising network learns to recover the clean image from the image of each step. We propose a dual-path guided model architecture BAF-UNet, as shown in Fig. 4. The first path processes the noisy input $\mathcal{X}_t'$, incorporating time-step embedding via sine-cosine positional encoding. The second path takes concatenated LR-HSI and HR-MSI as input, focusing



(a) Bilateral Attention Fusion UNet (BAF-UNet)

(b) Multi-scale Group Attention Block (MSGAB)

Figure 4. Architecture of the proposed BAF-UNet. Multi-scale supervision calculates the L1 loss between the result of fusing O_k with $\mathcal{Y}_{\uparrow n}$ and $\mathcal{Z}_{\downarrow n'}$, and the downsampled version of the real HR-HSI at the same resolution.

on structural and spectral priors. The two paths fuse features in deep layers rather than early concatenation, preventing information interference and enabling separate learning of noise distribution and structural features.

Multi-Scale Group Attention Block (MSGAB). It is the main block in BAF-UNet, and is a multi-branch module integrating group convolution, attention, and MLP to collaboratively capture local details, channel dependencies, and spatial context. First, the input In is first processed by feature extraction to obtain I, and then enhanced via spatial attention, which can be expressed as

$$I = \text{LReLU}(\text{LN}(\text{Conv}(In))), \tag{11}$$
$$J = \text{SpatialAttn}(I) \odot I, \tag{12}$$

where LReLU, LN, and $\odot$ denote LeakyReLU, LayerNorm, and element-wise multiplication, respectively.

Then, two parallel group convolutions extract complementary features. One branch employs 1 group convolution to obtain the feature g_1, for enabling dense cross-channel interaction, while another uses S group convolution to obtain the feature g_2 for maintaining sparse channel connectivity and preserving spectral characteristics.

$$g_1 = f^1_{\text{group}}(J), g_2 = f^S_{\text{group}}(J), \tag{13}$$

where f_{group} contains two 3×3 convolutions with different groups, two LeakyReLUs, and a LayerNorm.

Next, the weights β_i are obtained and weighted on the features of each branch to adaptively fuse global and local features, which can be expressed as

$$\beta_i = \text{Softmax}\left(\text{Conv}\left(\text{GAP}\left(g_i\right)\right)\right), \tag{14}$$

$$h = \sum_{i=1}^{2} \beta_i \cdot g_i, i = 1, 2 \tag{15}$$

where GAP is the global average pooling.

Finally, the fused features are nonlinearly transformed and enhanced by the MLP to obtain feature V, which is added to the input through a residual connection to produce the output. It can be expressed as

$$V = \mathbf{MLP}(h), Out = V + In. \tag{16}$$

Multi-scale Supervision. To fully exploit the information from LR-HSI and HR-MSI, we introduce a multi-scale supervision. Specifically, LR-HSI is upsampled using bicubic interpolation at different magnifications, while HR-MSI is downsampled to the corresponding resolution. Then, the concatenations are injected into each upsampling stage. Additionally, we employ progressively downsampled versions of the HR-HSI and Pseudo-MSI (denoted $\mathcal{X}'_{\downarrow n'}$) as per-stage supervision to explicitly constrain the upsampling process. It allows LR-HSI to guide spectral recovery and HR-MSI to refine spatial structure, simulating a super-resolution mapping that coordinates spectral and spatial optimization. The multi-scale L_1 loss function can be expressed as

$$L_{multi} = \sum_{k=0}^{3} \|O_k + \mathcal{Y}_{\uparrow n} \copyright \mathcal{Z}_{\downarrow n'} - \mathcal{X}'_{\downarrow n'}\|_1, \tag{17}$$

where $n = 2^k$, $n' = 2^{3-k}$, $\mathcal{Y}_{\uparrow n}$ denotes the image upsampled n times using bicubic interpolation for LR-HSI, $\mathcal{Z}_{\downarrow n'}$ represents an image upsampled n' times from HR-MSI, O_k is the output image of the k^{th} stage of BAF-UNet, and $\mathcal{X}'_{\downarrow n'}$ represents $\mathcal{X}'_0$ downsampled by n' times, respectively.

4. Experimental Results

4.1. Experimental Setup

Dataset Setting. To validate the superiority of our method, we conduct comparative experiments on the ICVL [1], Harvard [4], and Chikusei [44] datasets. The ICVL and Harvard datasets are processed by cropping $256 \times 256 \times 31$ patches. The Chikusei dataset is horizontally divided into two distinct parts. The upper and lower parts are separately tiled into non-overlapped $256 \times 256 \times 128$ patches, which are used for training and testing, respectively. LR-HSI is obtained by blurring with a 3×3 Gaussian filter (standard

Table 1. All methods are compared quantitatively on the ICVL, Harvard, and Chikusei datasets. ↑ and ↓ indicate that higher or lower values correspond to better results. The best results are highlighted in **bold**.

Methods	ICVL				Harvard				Chikusei			
	PSNR↑	SSIM↑	SAM↓	ERGAS↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓
CNMF [46]	36.21	0.9895	0.0512	2.2128	32.94	0.9697	0.0730	4.3996	32.45	0.9798	0.2936	5.7328
Hysure [32]	38.37	0.9901	0.0457	1.9432	32.65	0.9737	0.1177	5.2153	33.59	0.9802	0.2812	5.6064
LAGConv [17]	53.53	0.9996	0.0044	0.1846	47.75	0.9984	0.0265	1.0045	43.96	0.9960	0.1160	2.1326
DHIF [16]	52.57	0.9995	0.0042	0.1938	47.78	0.9973	0.0260	0.8087	44.06	0.9959	0.1173	2.0143
PSRT [5]	52.00	0.9987	0.0061	0.3669	46.76	0.9972	0.0244	1.0380	44.74	0.9968	0.1055	1.8804
DSPNet [34]	55.19	0.9996	0.0042	0.1609	48.68	0.9986	0.0237	0.8814	46.97	0.9979	0.0977	1.5178
PLR [31]	45.94	0.9854	0.0233	1.4218	41.02	0.9920	0.0653	1.9737	38.47	0.9826	0.1288	4.0849
SDP [26]	46.11	0.9913	0.0219	1.0692	42.31	0.9931	0.0678	1.7294	39.01	0.9842	0.1243	3.4658
ARGS [60]	46.56	0.9922	0.0210	0.6728	41.97	0.9938	0.0662	1.7595	39.43	0.9850	0.1224	3.2289
SMGU [43]	51.44	0.9985	0.0062	0.3921	46.77	0.9979	0.0280	1.1435	44.53	0.9961	0.1042	1.9832
LRTN [27]	52.35	0.9992	0.0051	0.3796	46.80	0.9974	0.0277	0.9837	43.77	0.9961	0.1129	1.9872
EMR-Diff	**55.40**	**0.9997**	**0.0040**	**0.1588**	**49.28**	**0.9990**	**0.0233**	**0.7800**	**47.55**	**0.9980**	**0.0950**	**1.4943**

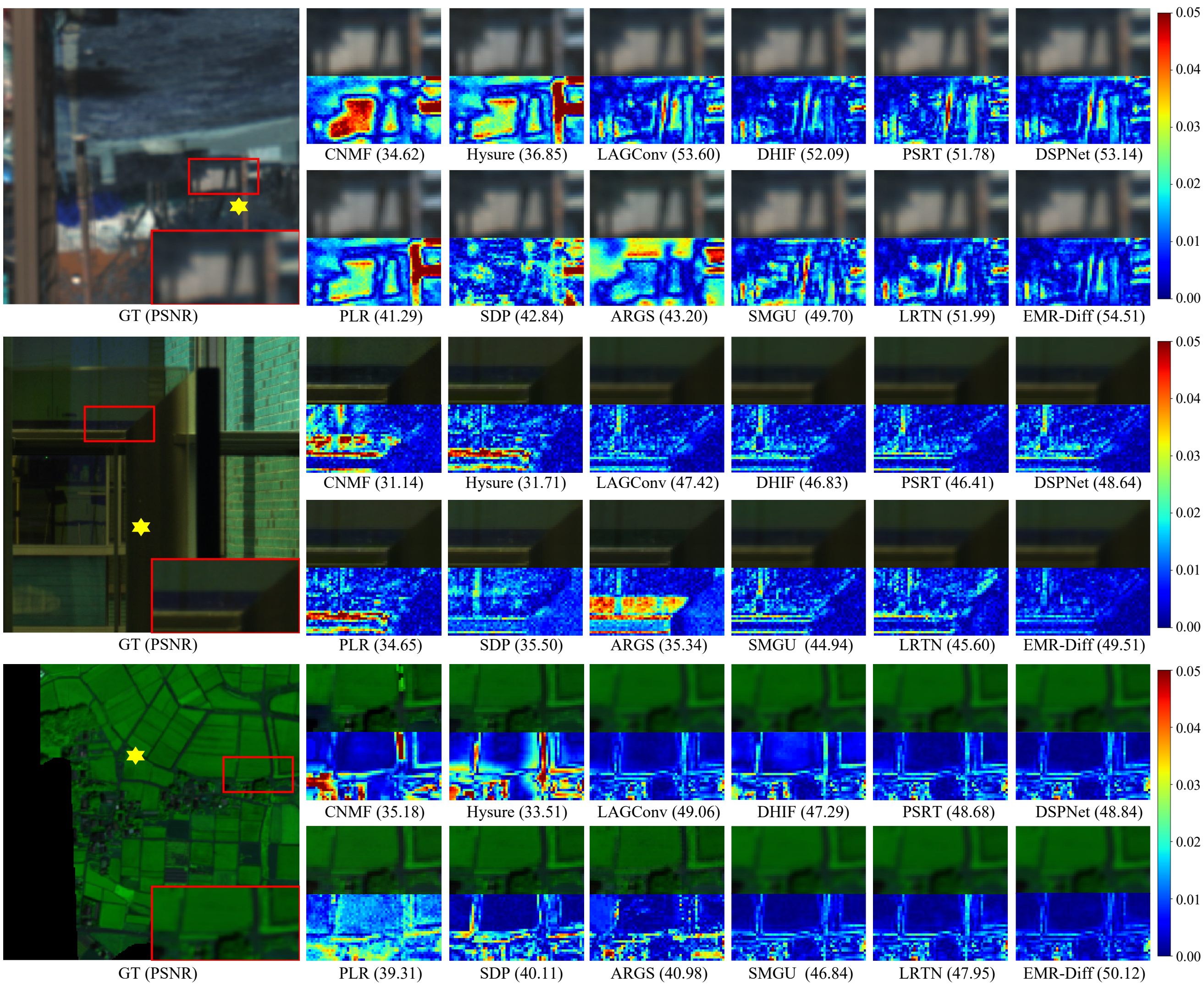


Figure 5. Visual comparison results for all methods across the ICVL, Harvard, and Chikusei datasets. Under the ICVL and Harvard datasets, the 10th, 20th, and 30th bands are selected to fuse into pseudo-color images, and the 30th band is used as the error map. In the Chikusei dataset, the 10th, 60th, and 80th bands are selected to fuse into pseudo-color images, and the 60th band is used as the error map.

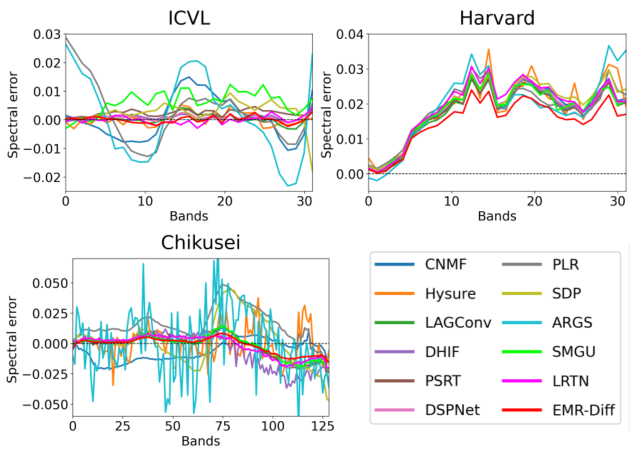


Figure 6. Spectral errors in the ICVL, Harvard, and Chikusei datasets. The comparison pixel is at the yellow star in Fig. 5.

deviation of 2) and downsampling by 8 times bicubic interpolation. HR-MSI is obtained by converting HR-HSI using the spectral response function of Nikon D700 camera.

Evaluation Metrics. To evaluate image quality, we use four metrics, including peak signal-to-noise ratio (PSNR), structural similarity (SSIM), spectral angle mapping (SAM), and relative dimensionless global error in synthesis (ERGAS). In general, lower ERGAS and SAM values and higher PSNR and SSIM values are better for fusion.

Comparison Methods. We conduct comprehensive comparisons with 10 state-of-the-art methods on three benchmark datasets. The competing methods are categorized into three groups, including traditional methods (CNMF [46], Hysure [32]), unsupervised learning methods (PLR [31], SDP [26] and ARGS [60]), and supervised learning methods (LAGConv [17], DHIF [16], PSRT [5], DSPNet [34], SMGU [43], and LRTN [27]). To ensure fairness, all methods are retrained on NVIDIA GeForce GTX 5070 Ti.

4.2. Quantitative and Qualitative Comparisons

Quantitative Results. The quantitative evaluation results across the three benchmark datasets are summarized in Table 1. EMR-Diff consistently achieves the best performance on all four metrics. Notably, traditional methods (CNMF, Hysure) have significant performance gaps compared to deep learning methods, which underscores the limitation of hand-crafted priors in modeling complex real-world scenes. Unsupervised learning methods (PLR, SDP, ARGS) possess better generalization, but their reconstruction quality is still limited. They often produce overly smooth outputs, which is evidenced by their lower PSNR and SSIM scores compared to supervised methods. Compared with supervised methods (LAGConv, DHIF, PSRT, DSPNet, SMGU, LRTN, EMR-Diff), our EMR-Diff method, with multimodal residual, edge-aware noise, and BAF-UNet architecture, can more effectively capture the data distribution of HR-HSI, achieving the best performance in all quantitative metrics. It demonstrates the effectiveness of the proposed method.

Table 2. Ablation study of multimodal residual.

Methods	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
None Residual	47.93	0.9984	0.0250	0.8994
Unimodal Residual	48.46	0.9986	0.0241	0.8025
Multimodal Residual	**49.28**	**0.9990**	**0.0233**	**0.7800**

Table 3. Ablation study of edge-aware noise.

Methods	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
Pure Noise	48.36	0.9986	0.0245	0.8198
Edge-aware Noise	**49.28**	**0.9990**	**0.0233**	**0.7800**

Table 4. Ablation study on BAF-UNet. MSS denotes multi-scale supervision. BAF-UNet(S) is with single supervision.

Module	UNet	B-UNet	AF-UNet	BAF-UNet(S)	BAF-UNet
Bilateral		✓		✓	✓
MSGAB			✓	✓	✓
MSS		✓	✓		✓
PSNR ↑	44.87	48.22	48.63	48.54	**49.28**
SSIM ↑	0.9972	0.9983	0.9986	0.9985	**0.9990**
SAM ↓	0.0260	0.0244	0.0237	0.0239	**0.0233**
ERGAS ↓	0.9369	0.8637	0.8024	0.8811	**0.7800**

Qualitative Visualization. The spatial visualization is shown in Fig. 5, and zoom-in image and the corresponding error map of each method are provided. The fused pseudo-color images produced by EMR-Diff are visually sharper and more similar to the Ground Truth (GT). The error map of EMR-Diff is noticeably bluer, while other methods show pronounced errors. It visually confirms that our edge-aware strategy and multimodal residual transfer effectively guide the model to accurately reconstruct critical spatial details. In Fig. 6, our EMR-Diff consistently maintains the lowest error curve and the smallest fluctuation range across the entire spectral range, indicating its ability to keep low spectral distortion in all bands. It further validates the effectiveness of the method we proposed in spectral information recovery.

4.3. Ablation Studies

Effect of Multimodal Residual. To demonstrate the efficiency of propagating multimodal residual through Markov chains, we compare the performance of three configurations, controlling all other experimental conditions to be the same, *i.e.*, no residuals, only the unimodal residual between LR-HSI and HR-HSI, and multimodal residual. As shown in Table 2, the introduction of residual information into the model can significantly improve performance, and our proposed multimodal residual mechanism achieves optimal results on all evaluation metrics. Compared to the conditions without residual and unimodal residual, multimodal resid-

Table 5. Ablation study of diffusion steps.

Steps	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
3	48.54	0.9987	0.0239	0.8009
4	48.70	0.9988	0.0237	0.7890
5	**49.28**	**0.9990**	**0.0233**	**0.7800**
10	48.89	0.9988	0.0236	0.7882

Table 6. Ablation study of pseudo-MSI synthesis.

Bands	PSNR ↑	SSIM ↑	SAM ↓	ERGAS ↓
(29,30,31)	48.89	0.9988	0.0235	0.7892
(1,15,31)	49.02	0.9989	0.0236	0.7848
(1,2,3)	**49.28**	**0.9990**	**0.0233**	**0.7800**

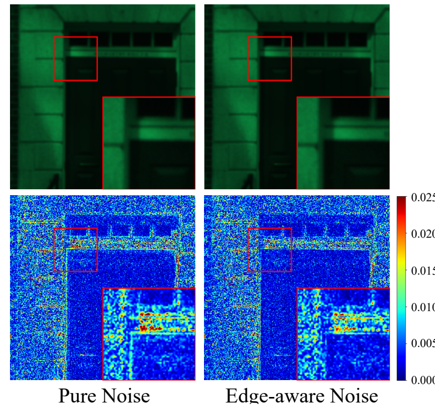


Figure 7. Visualization of pure noise and edge-aware noise.

ual improves PSNR by 1.35 dB and 0.82 dB, respectively. It demonstrates that transferring multimodal residual containing rich high-frequency details and spectral differences through Markov chains can effectively guide the model to recover accurate spatial structure and spectral properties.

Effect of Edge-aware Noise. To verify the role of edge-aware noise in reconstructing high-frequency details, we compared it with a baseline using pure Gaussian noise. As shown in Table 3, all metrics showed consistent improvement after adopting edge-aware noise, with PSNR increasing by 0.92 dB. As the error map visualization in Fig. 7 shows, images generated using edge-aware noise exhibit significantly smaller errors in edge regions. It indicates that by concentrating noise energy in high-frequency edge areas, this strategy successfully guides denoising network to prioritize restoring image details and textures crucial for visual perception, thereby significantly enhancing spatial fidelity.

Effect of BAF-UNet. We perform a modular ablation study on the key design of BAF-UNet, and the results are summarized in Table 4. Replacing the MSGAB module with a regular residual block in B-UNet leads to a significant performance degradation, with a PSNR drop of 1.06 dB. This demonstrates the powerful ability of MSGAB to collaboratively capture local details and channel dependencies by integrating grouped convolutions, spatial attention, and adaptive fusion. Compared to BAF-UNet, the early fusion AF-UNet results in a 0.65 dB decrease in the PSNR metric, which validates that the bilateral path design effectively separates the denoising and guidance tasks of noisy input and observed input, avoiding feature confusion and enabling the network to focus on learning specific features from different sources. Meanwhile, the BAF-UNet (S) using only single-scale supervision results in a PSNR drop of 0.74 dB, proving that the multi-scale supervision mechanism effectively addresses the blind upsampling issue and achieves synergistic optimization of spectral and spatial information.

Effect of Diffusion Steps. We further investigate the trade-off between reconstruction quality and efficiency in the inference process with respect to the number of diffusion steps. As shown in Table 5, when the number of steps is 5, EMR-Diff achieves the best image quality. When the number of steps is further reduced to 4 or increased to 6, the performance degrades significantly. Thus, 5 steps are proven to be the optimal balance between image quality and inference efficiency, highlighting the substantial advantage of our method in sampling efficiency.

Effect of Pseudo-MSI Synthesis. We ablate different band selections for pseudo-MSI synthesis in Table 6. Using the first three bands (1,2,3) achieves the best performance, outperforming both uniformly sampled bands (1,15,31) and the last three bands (29,30,31). It confirms that the initial band typically corresponds to the visible spectrum, while the visible light region contains richer spatial details and higher frequency information, which is crucial for guiding diffusion models to restore structural content.

5. Conclusion

This paper proposes an Edge-aware Multimodal Residual Diffusion Model (EMR-Diff) for HSI super-resolution. It significantly shortens the diffusion process by embedding multimodal residual into Markov chains, achieving efficient and high-quality reconstruction within dozens of steps. Meanwhile, we innovatively utilize the edge information of HR-MSI to construct edge-aware noise, guiding the model to focus on the recovery of image high-frequency details during denoising. In addition, the designed BAF-UNet architecture fully explores and utilizes the spatial and spectral priors in the observed images through dual-path information decoupling, MSGAB module adaptive feature fusion, and multi-scale progressive supervision. Experimental results on various benchmark datasets show that the proposed method is superior to existing mainstream approaches in multiple quantitative metrics and visual quality. In the future, exploring more efficient samplers and how to enhance the model's generalization ability across sensors and scenes is a challenging task of significant practical importance.

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